$p_1 = K_H x_1$----------(1)

At pressure 6.0 atm,

$p_2 = K_H x_2$----------(2)

Dividing equation (1) by (2)

From equation $p_1/p_2 = x_1/x_2$

$1.5/6.0 = 0.24/x_2$

Therefore $x_2 = 0.24 \times 6.0/1.5 = 0.96$ g/L

Evaluate Yourself

7) Calculate the proportion of O_2 and N_2 dissolved in water at 298 K. When air containing 20% O_2 and 80% N_2 by volume is in equilibrium with water at 1 atm pressure. Henry's law constants for two gases are $K_H(O_2) = 4.6 \times 10^4$ atm and $K_H(N_2) = 8.5 \times 10^4$ atm.

8) Explain why the aquatic species are more comfortable in cold water during winter season rather than warm water during the summer.

9.6 Vapour pressure of liquid

Generally, liquids have a tendency to evaporate. If the kinetic energy of molecules in the liquid state overcomes the intermolecular force of attraction between them, then the molecules will escape from the liquid state. This process is called 'evaporation' and it happens on the surface of the liquid.

If evaporation is carried out in a closed container then the vapour remains in contact with the surface of the liquid. These vapour molecules are in continuous random motion during which they collide with each other and also with the walls of the container. As the collision is inelastic, they lose their energy and as result the vapour returns back to liquid state. This process is called as 'condensation'.

Evaporation and condensation are continuous processes. If the process is carried out in a closed system, a stage is reached when the rate of evaporation becomes equal to the rate of condensation. Thus, an equilibrium is established between liquid and its vapour. The pressure of the vapour in equilibrium with its liquid is called vapour pressure of the liquid at the given temperature. The vapour pressure of a liquid depends on its nature, temperature and the surface area. The following simple apparatus demonstrates the measurement of vapour pressure of a liquid.

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